Why Cells Are Small

Finding out what actually sets the size limit on a cell — and checking, number by number, whether the membrane ever runs short.

Name ______________________ Date ______________________ Period ______________________

Demonstration: andresdbp.github.io/resources/demos/why-cells-are-small/

1 Before you open anything

Answer from what you already know. The point is to have a number on paper that the demonstration can later confirm or refute.

PREDICTION

A spherical cell sits in fluid holding oxygen at 100 μM. Oxygen diffuses in through the membrane and is burned at a steady rate throughout the cytoplasm. At a radius of 60 μm, the concentration at the very centre works out to 88 μM.

Predict the centre concentration if the radius is doubled to 120 μm, with nothing else changed.

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Write one sentence saying what reasoning produced that number.

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2 Set the cell up and read it

Open the demonstration. The left side draws the cell and the oxygen inside it; the right side reports the same cell as numbers.

1. Press **Reset everything**. The *Sphere* tab is selected and the *Working muscle* preset is on.
2. Confirm the three sliders in *The physics* read **k = 40.0 μM s^{-1}**, **D = 2000 μm^2 s^{-1}** and **C_S = 100 μM**, and that **Radius R** reads **120 μm**.
3. Fill in the first row of the table from *The answer* panel and the *Does the membrane fail?* panel.
4. Now drag **Radius R** to **240 μm** and fill in the second row. Record the flux numbers with their units, which change as the cell grows.

Radius R	Oxygen at the centre	Critical size for this shape	Dead core radius	Volume that is anoxic	Oxygen consumed	Oxygen crossing the membrane
120 µm						
240 µm						

a. Compare the last two columns within each row. What is the relationship between oxygen consumed and oxygen crossing the membrane, and does it change when the cell grows past the critical size?

b. The 240 µm cell has a dead core. Given your answer to *a*, state whether the membrane failed to supply this cell, and say what did run out instead.

> The two flux numbers are computed independently: one from k × volume, the other from Fick's law at the membrane. They are not the same calculation printed twice.

3 How the centre falls as the cell grows

Keep the *Sphere* tab and the *Working muscle* settings from Part 2. Change only **Radius R**. The outside concentration stays at 100 µM, so the deficit is 100 minus whatever the centre reads.

1. Set **Radius R** to each value in the table and read **Oxygen at the centre**.
2. Fill in the deficit column yourself: 100 - (centre reading).
3. Fill in the last column: each deficit divided by the deficit in the first row.

Radius R	R ÷ 30 µm	Oxygen at the centre (µM)	Deficit = 100 - centre (µM)	Deficit ÷ deficit in row 1
30.0 µm	1			1
60.0 µm	2			
90.0 µm	3			
120 µm	4			
150 µm	5			

a. Compare column 2 with column 5. Complete the rule in your own words: multiplying the radius by *m* changes the drop in concentration between the membrane and the centre by a factor of __________.

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b. Use your rule — not the demonstration — to work out the radius at which the deficit would reach the full 100 μM, so the centre reaches zero. Show the arithmetic, then check it against the **Critical size for this shape** readout.

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c. Turn back to your Part 1 prediction. The table contains both answers. Was your number right? If it was not, name the specific assumption in your reasoning that the data contradicts.

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4 Two ways of using this

The first task is the comparison the demonstration is built around. The second is a shortcut that looks like it should work.

A. Same cytoplasm, different shape

1. Press **Reset everything**, then set **Radius R** to **250 µm**. Fill in the sphere row.
2. Tick **Keep the volume fixed when I change shape**.
3. Press the **Flat sheet** tab. The dimension control is now *Half-thickness L* and the demonstration has re-solved it to hold the same cytoplasm. Fill in the sheet row.

Shape	Radius R / Half-thickness L	Volume (μm^3)	SA : V (μm^{-1})	Oxygen at the centre	Volume that is anoxic
Sphere	250 µm				
Flat sheet					

a. By what factor did SA:V rise? By what factor did the distance from the membrane to the deepest point fall? Show both.

b. The centre went from anoxic to nearly the full outside value. Using the two factors from *a*, argue which of them can account for that change and which cannot.

c. The geometry constant n is 6 for a sphere and 2 for a sheet. Part of the improvement in *b* comes from the distance and part is given back by n. Which way does the change in n push the centre concentration, and why does the sheet still win overall?

B. Where the SA:V shortcut fails

1. Press **Reset everything**. Leave **Keep the volume fixed** *unticked*.

2. Press the **Flat sheet** tab and set **Half-thickness L** to **50.0 µm**. Fill in the first row.

3. Change nothing else except **Elongation / flattening**, from **20×** to **100×**. Fill in the second row.

Elongation	Volume (μm^3)	Surface area (μm^2)	SA : V (μm^{-1})	Oxygen at the centre
20×				
100×				

d. Volume and surface area both grew by more than twenty-fold, and SA:V moved slightly in the direction that is supposed to mean trouble. What happened to the centre concentration?

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e. A prediction written as “centre concentration follows SA:V” would fail here. State the quantity that the centre concentration does follow, and explain why a sheet can be widened without limit without the middle noticing.

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5 Solve these without the demonstration

Close the page, or look away from it. Show your working. Use $C(\text{centre}) = C_S - k\,a^2 / (nD)$ and $a_{crit} = \sqrt{(nDC_S/k)}$, with $n = 6$ for a sphere, 4 for a cylinder and 2 for a sheet. Then reopen the demonstration, set the same conditions, and check yourself.

1. Resting tissue: $k = 5\ \mu M\ s^{-1}$, $D = 2000\ \mu m^2\ s^{-1}$, $C_S = 60\ \mu M$. For a spherical cell of radius 200 μm, find the oxygen concentration at the centre, and find the critical radius. State whether this cell has a dead core.

2. Working muscle in arterial blood: $k = 40\ \mu M\ s^{-1}$, $D = 2000\ \mu m^2\ s^{-1}$, $C_S = 130\ \mu M$. Find the largest radius a spherical cell can have and still have oxygen everywhere inside.

3. Keep the values from problem 2 and switch to a flat sheet. Find the critical half-thickness. It comes out smaller than the sphere's critical radius. Explain why that is not a contradiction of the claim that flat cells can grow larger than spherical ones — your answer needs to say what is being held fixed in each comparison.

4. A spherical cell with $k = 40\ \mu M\ s^{-1}$, $D = 2000\ \mu m^2\ s^{-1}$ and $C_S = 100\ \mu M$ grows from a radius of 40 μm to 80 μm. Show that its SA:V is exactly halved, then find the centre concentration at both radii. Explain why the drop from the membrane to the centre does not change by the same factor as SA:V, and what that rules out as the cause of the size limit.

6 On your own

Nothing here is graded. The demonstration does considerably more than this worksheet uses.

- Press **A large protein** and read the *How long to reach this steady state* panel. This is the one setting where the transient time becomes a serious constraint. Work out what had to change to get there, and why the same argument does not apply to oxygen.
- Use the **Cylinder** tab with **Keep the volume fixed** ticked, and see where a cylinder lands between the sphere and the sheet in the *Same cytoplasm, three shapes* table. Its geometry constant is 4. Why is it between the other two rather than outside them?
- Hold the radius fixed and walk **Outside concentration C_S** from 130 μM, roughly arterial, down to 50 μM, roughly the venous end of a capillary bed. Watch the critical size. What does this predict about where in a tissue hypoxia appears first?
- Read *Where this model simplifies*. Pick the assumption you think a genuinely large cell is most likely to break, and say what it would gain by breaking it.